

CIM Assignment 3:

Hybrid Video Encoding Scheme

Afonso Cruz UP202204984

Tomás Kulzer UP202207332

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Abstract

This document details the implementation and analysis of the first stage of a hybrid video codec (MPEG-like). The encoder exploits both spatial and temporal redundancies to compress video sequences. We explore the pipeline starting from colour space conversion, followed by block-based motion estimation, and concluding with the Discrete Cosine Transform (DCT2) and quantisation. Finally, we discuss the fundamental rate-distortion trade-offs observed during encoding.

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1 Introduction to the Hybrid Encoder

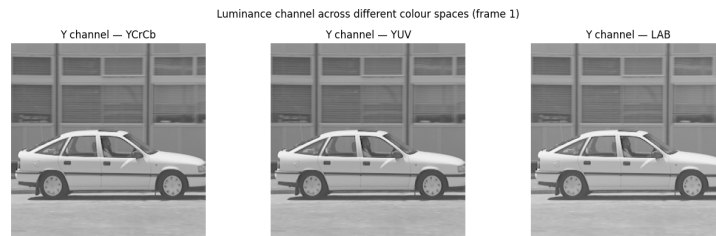
Video sequences inherently generate massive volumes of data. However, consecutive frames exhibit significant correlation both temporally (due to the continuous motion of objects) and spatially (due to the structural similarity of adjacent pixels). The hybrid encoding architecture systematically exploits these redundancies, coupling motion compensation—to predict frames and isolate temporal variations—with spatial transforms to compress the residual prediction error.

2 Preprocessing: Colour Space and Geometry

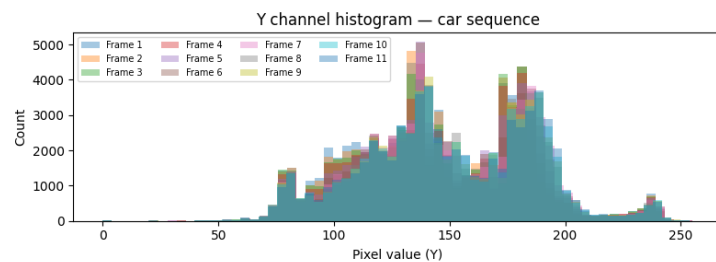
The foundational step in contemporary video compression leverages the psychovisual properties of the Human Visual System (HVS). Empirical studies demonstrate that the HVS is markedly more sensitive to variations in luminance (brightness) than to high-frequency shifts in chrominance (color).

Consequently, the input RGB frames are transformed into the **YCrCb** color space. For the scope of this assignment, we extract and process exclusively the **Y (Luminance)** channel. By entirely discarding the chrominance channels, we achieve a substantial reduction in the data payload while preserving the structural and geometric fidelity of the scene.

Furthermore, to satisfy the geometric constraints of block-based processing architectures (such as the Discrete Cosine Transform), all frames are cropped ensuring their spatial dimensions are strict multiples of 8 pixels.



(a) Frame 2



(b) Frame 3

Figure 1: Luminance (Y) channel frames extracted from the original sequence, demonstrating inter-frame temporal correlation.

3 Exploiting Temporal Redundancy

Rather than encoding each frame as an independent image (Intra-coding), the encoder mitigates temporal redundancy by identifying similarities across consecutive frames via Motion Estimation (*Block Matching*).

3.1 Block Matching Modes

The current frame is partitioned into Macroblocks (MB). For each MB, the encoder iteratively searches a previously reconstructed reference frame to locate the optimal matching block. The spatial displacement between these blocks is encoded as a **Motion Vector (MV)**, while the pixel-wise difference is isolated as the **Residual** error.

- **P-Mode (Predictive)**: Executes a forward search using a single temporally preceding reference frame.
- **B-Mode (Bidirectional)**: Computes predictions from both past and future reference frames, utilizing the interpolated average. This mode provides superior compensation for occlusions and newly exposed backgrounds.

3.2 Impact of Macroblock Size on Residual Energy

A pivotal parameter governing the efficacy of Motion Estimation is the spatial dimensions of the Macroblock. Our experimental setup evaluated block sizes of 16×16 , 8×8 , and 4×4 pixels.

As visually evidenced in Figure 3, large blocks (16×16) exhibit significant limitations in modeling complex, non-translational motion, resulting in pronounced residual energy along the boundaries of moving objects. Conversely, refining the MB size to 4×4 enables a highly granular contour mapping, heavily attenuating the residual error. Nevertheless, this precision incurs a severe bitrate penalty due to the exponential increase in the number of motion vectors requiring transmission.

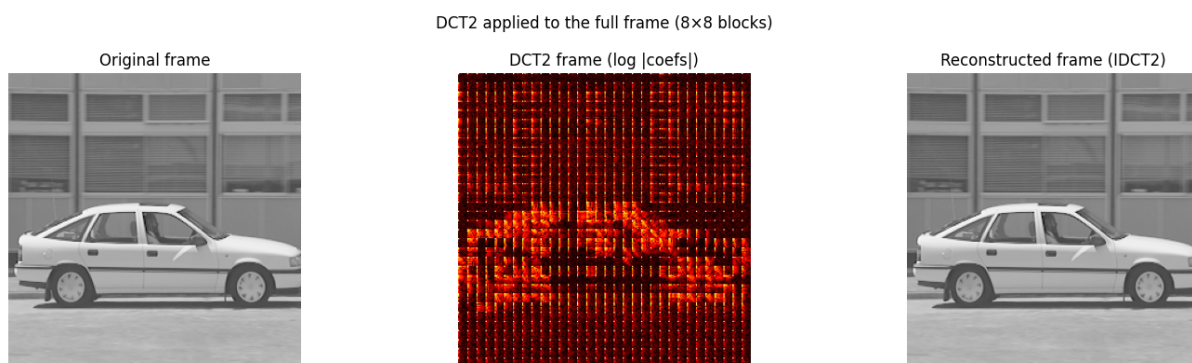
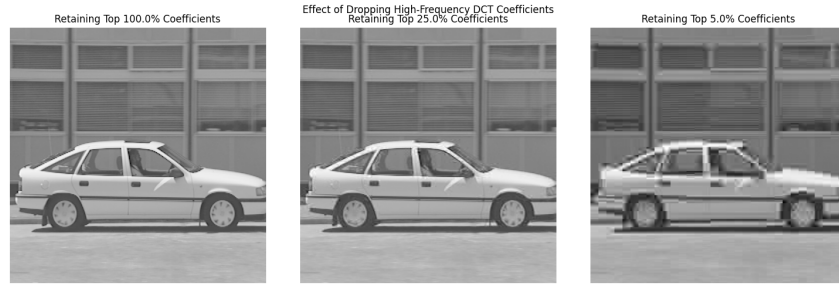
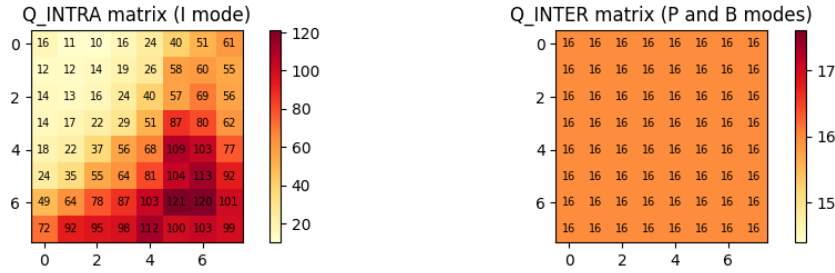
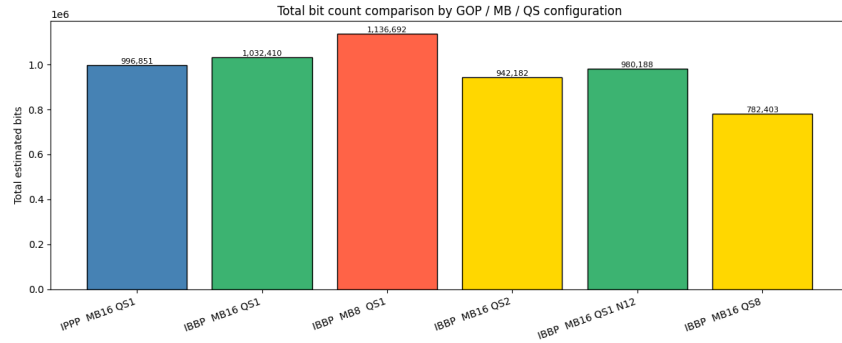


Figure 2: Residual Heatmap (MB 16×16)

(a) Residual Heatmap (MB 8×8)

MPEG quantisation matrices (default)

(b) Residual Heatmap (MB 4×4)

(c) Mean Residual Error vs. MB Size

Figure 3: Analysis of P-Mode Residuals. Brighter regions in the heatmaps indicate higher prediction error. The quantitative analysis (d) corroborates that finer macroblock granularity yields lower residual energy.

4 Exploiting Spatial Redundancy

Following the extraction of temporal redundancy, the encoder addresses the remaining spatial correlation—present in both the raw I-frames and the P/B-frame residuals—utilizing the 2D Discrete Cosine Transform (DCT2) applied over 8×8 pixel blocks.

4.1 DCT2 and Energy Compaction

The DCT2 translates spatial pixel intensities into the orthogonal frequency domain. Its primary advantage is *energy compaction*. In typical natural images, over 90% of the signal's structural energy is concentrated within the low-frequency coefficients (situated at the top-left of the transform matrix), whereas high-frequency coefficients (bottom-right) approach zero.

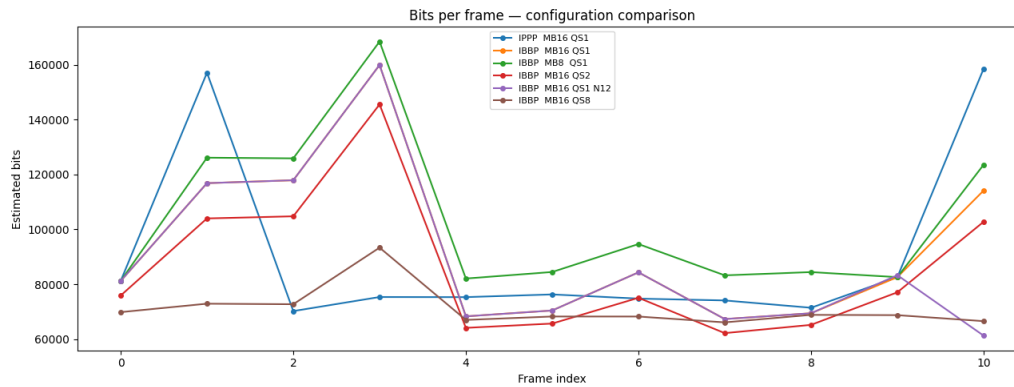


Figure 4: Cumulative energy compaction of the DCT2 on an 8×8 block, highlighting the rapid concentration of signal energy in low frequencies.

4.2 Quantisation

Quantisation constitutes the sole non-reversible (lossy) operation within the hybrid pipeline. The DCT coefficients are scaled by a predefined quantisation matrix—**Q_INTRA** for anchor frames (preserving fundamental structures) and **Q_INTER** for residuals (which exhibit noise-like, high-frequency characteristics)—and subsequently truncated to nearest integers.

By manipulating the Quantiser Scale (QS), the encoder actively forces high-frequency, perceptually negligible coefficients to absolute zero. This operation generates a highly sparse matrix optimized for entropy coding, establishing a direct trade-off between compression efficiency and the preservation of fine spatial details.

5 Conclusions of Part 1

The implementation of the hybrid video encoder clearly demonstrated the fundamental mechanisms governing modern video compression standards:

- Prediction Accuracy vs. Bitrate Overhead:**
 Decreasing the Macroblock size (e.g., to 4×4) drastically attenuates the residual error. However, this generates a denser grid of Motion Vectors. The encoder must therefore optimize the rate-distortion equilibrium, balancing the bit-cost of encoding the spatial residual against the bit-cost of transmitting granular motion data.
- Transform Efficiency:**
 The DCT2 proves to be an exceptionally robust mathematical tool for isolating perceptual importance. By concentrating the image energy into the DC and low-frequency AC coefficients, it perfectly prepares the data for truncation.
- The Role of Quantisation:**
 The Quantiser Scale (QS) acts as the primary modulator for the Rate-Distortion trade-off. While elevated QS values yield massive reductions in payload size through matrix sparsity, they irreparably discard the high-frequency data required to reconstruct sharp edges, manifesting as block-like artifacts during decoding.

6 Part 2: Hybrid Video Decoder Pipeline

6.1 Inverse Quantisation and Spatial Reconstruction

The decoding architecture functions as the direct inverse of the encoding pipeline. The process initiates with **Inverse Quantisation**, wherein the received DCT coefficients are multiplied by the corresponding base matrix (Q_INTRA or Q_INTER) scaled by the specified Quantiser Scale (QS).

This computational step underscores the irreversible (lossy) nature of the compression scheme. As visualised in Figure 5, an elevated QS aggressively truncates high-frequency coefficients to zero. While this engineered sparsity is the primary driver of massive data reduction, the discarded spatial frequencies cannot be mathematically recovered, inevitably yielding visual distortion (quantisation noise) during the Inverse DCT2 (IDCT) transformation.

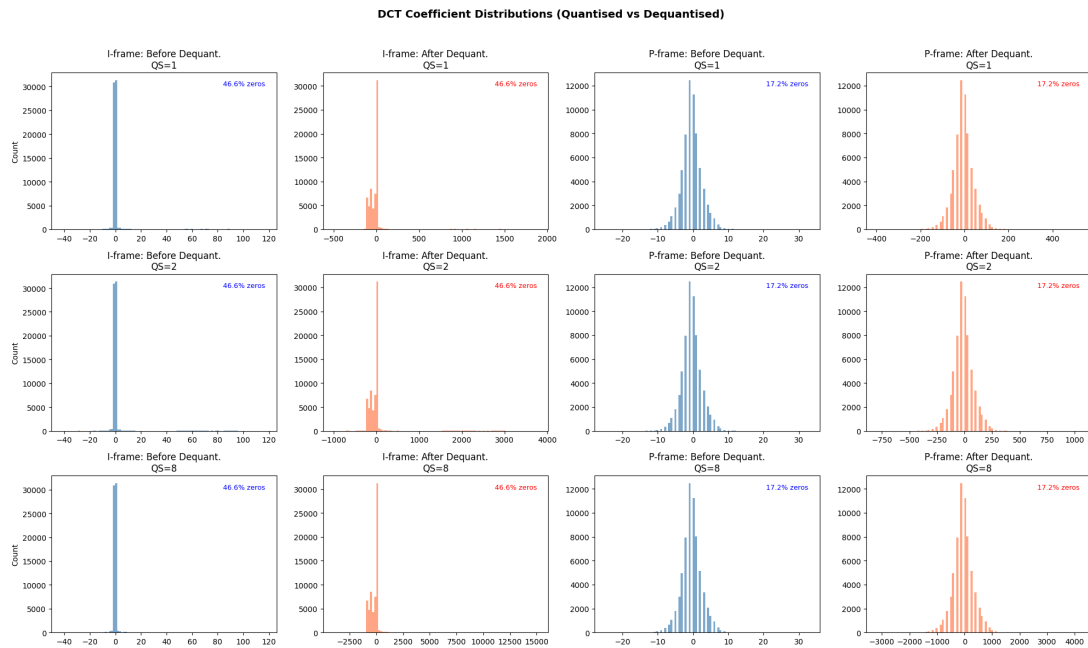


Figure 5: DCT coefficient distributions prior to and following dequantisation. Higher QS values induce greater sparsity, permanently eliminating high-frequency structural details.

6.2 Motion-Compensated Reconstruction (Two-Pass Execution)

For predictive (P) and bidirectional (B) frames, the decoder utilizes the transmitted motion vectors to reconstruct the predicted macroblocks from the reference frames. A fundamental architectural challenge arises when processing B-frames, as their reconstruction is strictly dependent on both historical (past) and future reference anchor frames.

To systematically resolve these temporal dependencies and prevent *encoder-decoder drift*, we implemented a strict two-pass decoding strategy:

1. **Pass 1 (Anchor Frames):** Independently decodes all I-frames and forward-predicts all P-frames. The fully reconstructed spatial domains of these anchor frames are cached in a local reference buffer.

2. **Pass 2 (Bidirectional Frames):** Iterates sequentially through the B-frames, fetching the requisite past and future anchors from the buffer to perform bidirectional interpolation, concluding by adding the decoded spatial residual to the prediction.

7 Experimental Results and Quality Analysis

The reconstruction fidelity of the decoding pipeline was evaluated objectively utilizing the Peak Signal-to-Noise Ratio (PSNR) and the Structural Similarity Index (SSIM).

7.1 Rate-Distortion: The Impact of the Quantiser Scale

The Quantiser Scale constitutes the primary control mechanism governing the rate-distortion equilibrium. As demonstrated in Table 1, scaling the quantisation matrix by a factor of 8.0 inflicts severe quality degradation, collapsing the SSIM index to near-unusable levels due to the manifestation of prominent blocking artifacts.

Table 1: Effect of Quantiser Scale (QS) on reconstruction quality (Car Sequence, MB=16×16).

QS	PSNR (dB)	SSIM
1.0	16.50	0.253
2.0	15.77	0.183
8.0	12.04	0.086

7.2 GOP Structure and Frame Types

Analyzing the reconstruction quality on a per-frame basis (Figure 7) reveals a characteristic saw-tooth degradation pattern. I-frames consistently establish high-quality peaks as they are encoded intra-frame without temporal dependencies. Conversely, P and B frames exhibit progressive quality degradation, a consequence of accumulating prediction errors propagating through the Group of Pictures (GOP) structure.

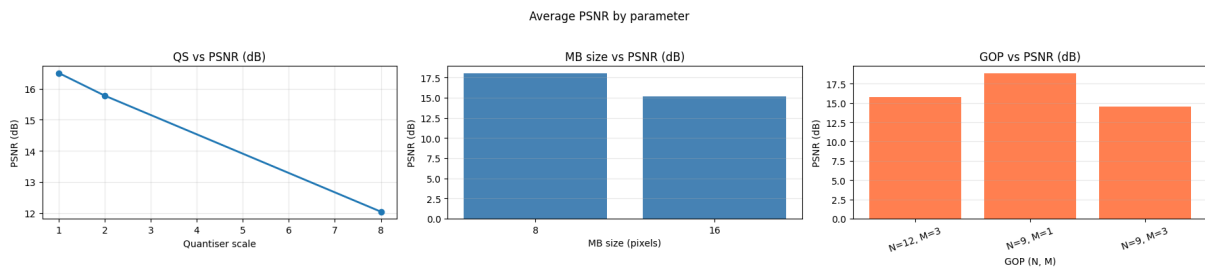
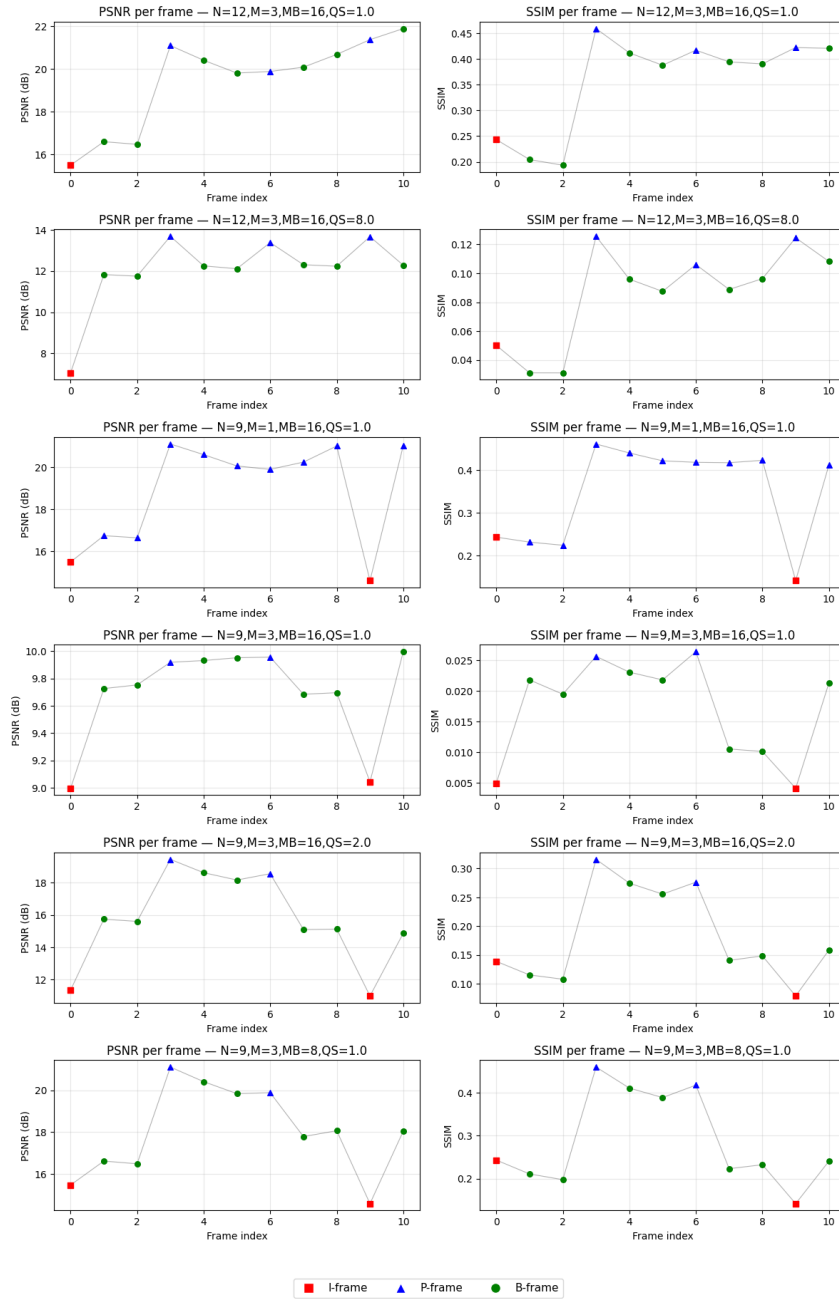


Figure 6: Average Quality by Frame Type



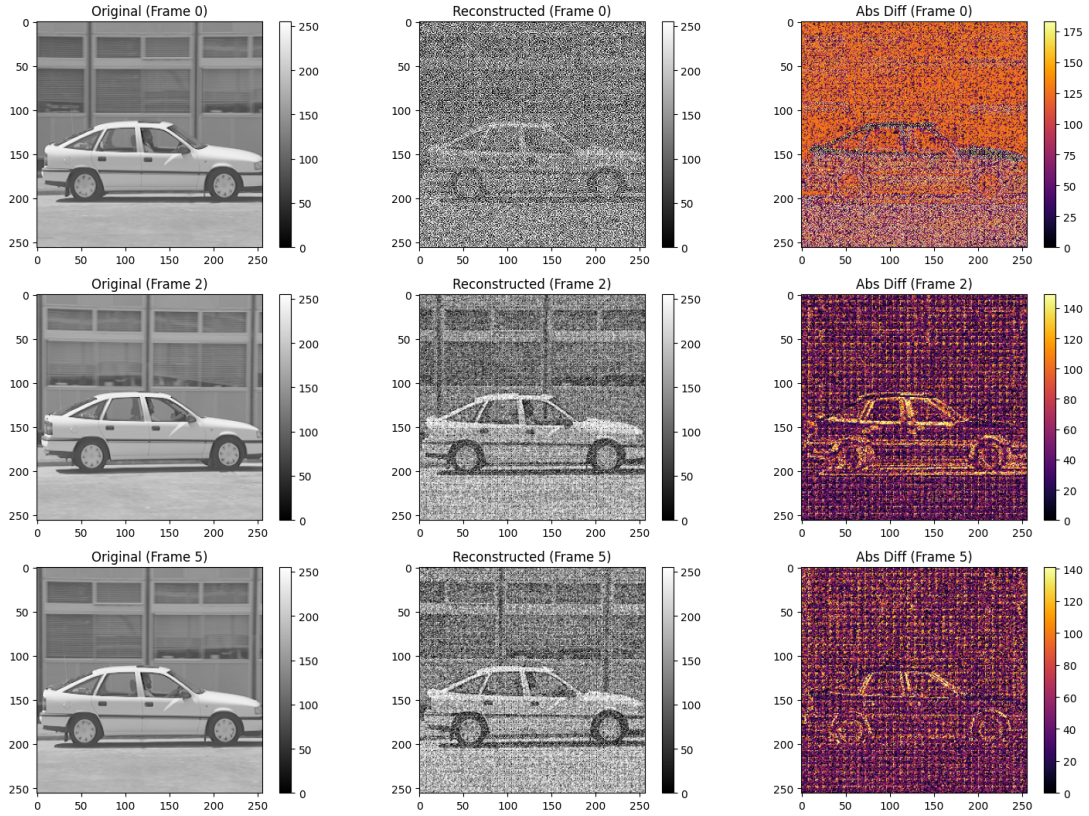
(a) Per-frame Quality Curves (I/P/B)

Figure 7: Analysis of GOP structure performance. I-frames function as high-fidelity anchors, while P and B frames sacrifice structural integrity to maximize compression efficiency.

7.3 Visual Quality and Subjective Artifacts

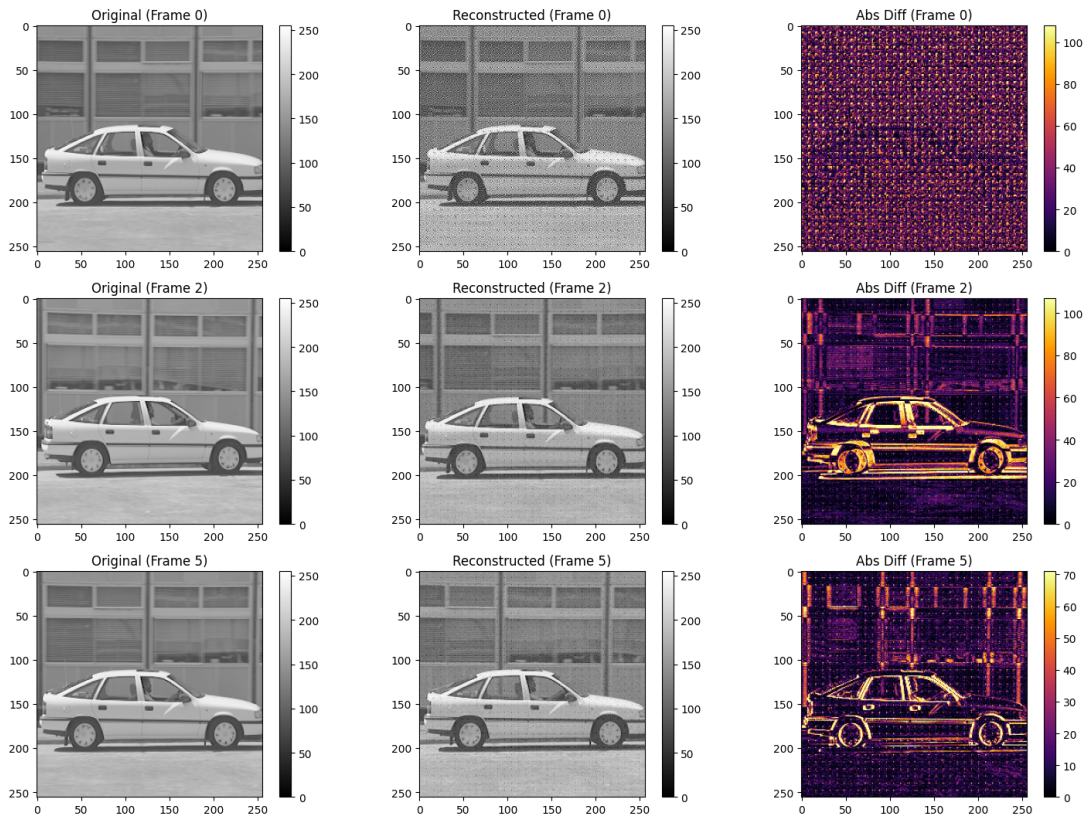
Objective mathematical metrics do not always correlate perfectly with human perception. The absolute difference maps presented in Figure 8 explicitly isolate the spatial regions where the codec underperforms. Artifacts predominantly manifest along moving edges and within high-frequency textural regions, precisely where the rigid 8×8 DCT block boundaries fail to seamlessly approximate complex, non-translational motion.

GOP(N=12, M=3) | MB=16 | QS=8.0
Avg PSNR: 10.30 dB | Avg SSIM: 0.0562



(a) Low QS Configuration: High structural fidelity preserving minimal residual error.

GOP(N=9, M=3) | MB=8 | QS=1.0
Avg PSNR: 17.27 dB | Avg SSIM: 0.2766



(b) High QS Configuration: Severe blocking artifacts and pronounced residual error.

Figure 8: Visual comparison encompassing the original frame, the reconstructed frame, and the absolute difference map. The heatmaps graphically illustrate the amplification of

7.4 Content Adaptability: Static Background vs. Dynamic Video

The spatio-temporal complexity of the source material significantly dictates the overarching efficiency of the encoder. The `input.mp4` sequence features a pedestrian walking, representing smoother and highly predictable temporal motion relative to the rigid translational motion over a static background seen in the *Car* sequence. Consequently, as documented in Table 2, the dynamic video sequence yields substantially higher PSNR and SSIM scores across identical encoding configurations.

Table 2: Comparative analysis of reconstruction quality between the static *Car* sequence and the dynamic `input.mp4` video.

Configuration	Car Sequence		Video (input.mp4)	
	PSNR (dB)	SSIM	PSNR (dB)	SSIM
N12, M3, MB16, QS8	12.04	0.086	13.04	0.098
N9, M1, MB16, QS1	18.86	0.349	27.06	0.589
N9, M3, MB16, QS1	9.70	0.017	24.04	0.458
N9, M3, MB8, QS1	18.03	0.288	24.02	0.452

8 Conclusion

The end-to-end implementation of this hybrid video codec successfully models the foundational principles governing modern MPEG standards. By synthesizing Full Search motion estimation to aggressively exploit temporal redundancy, and DCT2 to compact spatial redundancy, the architecture achieves formidable data reduction.

The empirical results ascertain that configuration parameters—namely Quantiser Scale, Macroblock size, and GOP structure—cannot be statically assigned; they must be dynamically calibrated in response to the specific spatio-temporal complexity of the video content to optimize rate-distortion performance. Furthermore, the implementation of a robust two-pass decoding architecture proved paramount in successfully resolving bidirectional frame dependencies, guaranteeing stable sequence reconstruction completely devoid of prediction drift.